What Have Been Driving India's Economic Growth? An Empirical Analysis

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Abstract

The present study is an attempt to identify the major sources of economic growth in India over the period 1971–2016 by employing the auto-regressive distributed lag (ARDL) bounds testing model. The long-run estimates of the ARDL model indicate that the economic growth in India is predominately driven by per capita capital and financial development while inflation retards economic growth. Contrary to the long run, the short-run results of error correction representation suggest that per capita capital along with human capital development has a significant positive impact on economic growth in India, while inflation curtails economic growth. India has some important policy insights to draw from these findings that the government policies in India need to emphasize the institutional mechanisms that further strengthen the Indian financial system which would increase its depth, scope and stability to foster economic growth. Further, the policymakers in India should strengthen their anti-inflationary measures, through supply-side reforms, to avoid the negative effects of inflation on economic growth. Finally, the government policies in India should also place a considerable emphasis on investment in human capital, which in turn fosters economic growth.

JEL Codes: O4, G2, E24, C1

Keywords

Economic growth, financial development index, inflation, principal component analysis, ARDL model

I. Introduction

Mankind has grown from innocence to strength by putting in consistent efforts to rearrange the existing resources and meticulously trying to add value to them. Thanks to human ingenuity and innovation that gave an impetus to this process, which resulted in richer and prosperous economies than they used to be earlier. However, all countries across the world did not grow in the same manner. The pattern, composition, magnitude and speed of growth in their respective economies differed over space and time. This

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divergence of growth rates among the countries gave rise to fundamental questions of why do economic growth rate in countries around the world change over time? And why do some countries grow faster than other countries?[1] While there have been rigorous attempts to find optimal answers to these questions, a consensus has not been arrived at, in the literature. For instance, there is a vast theoretical and empirical literature (Barro, 1991; Dollar, 1991; Lucas, 1988; Podrecca & Carmeci, 2001; Romer, 1986; Solow, 1956) on the issue of what makes a country's economy to grow. However, the findings of these studies are quite contrasting and divergent from each other. This might be due to the diverse nature of the principal determinants of economic growth and their variation across countries. Within the framework of economic growth theory, the neoclassical growth theory and the endogenous growth theory are two pertinent ideologies that have been largely, the subject of the debate on economic growth.

The Neoclassical theory, based on Solow's growth model, postulated that economic growth can be attributed to an accumulation of physical capital in the short run, while technological advancement, though regarded as an exogenous determinant, in the long run. In contrast to neoclassical growth theory, endogenous growth theory developed by Romer (1986) and Lucas (1988) advocates that the accumulation of knowledge, innovation, and public infrastructure will foster economic growth. In this context, the inclusion of human capital stock as one of the pertinent forces that drives economic growth, by complementing economic growth in the short run is the pertinent contribution of endogenous theory. This theory also made another key contribution, by including the productivity of factors like learning-by-doing and useful technological knowledge, in the long run. These theoretical advancements have, further, inspired a wide range of empirical studies to assess the impact of various macroeconomic variables on economic growth.

Thus, the macroeconomic factors such as gross fixed capital formation, that is, investment (Anwer & Sampath, 1999; Dollar, 1991; Podrecca & Carmeci, 2001), human capital (Barro, 1991; Hanushek & Wößmann, 2007; Pelinescu, 2015), inflation (Barro, 1991; Fischer, 1993; Gokal & Hanif, 2004), financial development (Cherif & Dreger, 2016; King & Levine, 1993; Murthy & Samantaraya, 2014; Rajan & Zingales 1998), an exchange rate (Alagidede & Ibrahim, 2017; Dollar, 1992; Rodrik, 2008) and trade openness (Awokuse, 2008; Dollar, 1992; Dollar & Kraay, 2002; Murthy et al., 2014) have occupied a central place in the growth literature.

However, despite the numerous studies, consensus on this issue of the relationship with economic growth is still a distant dream. Probably this may be due to the inadequate conceptualisation of the underlying process of economic performance. This is further, partly due to the lack of a generalised theory of economic growth. Therefore, the studies examined the determinants of economic growth so far, based on different theoretical frameworks, have often reported inconsistent findings on the source of economic growth, which in the end, have become a major challenge for policymakers in designing a universal policy.

Given this backdrop, it is pertinent to note that there is a clear shift in the locus of global economic gravity from the West to the Asian continent, and it appears that India occupies the centre stage now. This is evident from the report of the Centre for Economic and Business Research (2022), which predicts that India would be surpassing Germany by 2031 to become the third largest economy of the world with a GDP of nearly $6.8 trillion. For a country that reeled under the colonial rule and exploitation for nearly two centuries, and having inherited a poor and largely illiterate population at the time independence, this is a remarkable feat. It would be of great interest to understand what would have driven such an incredible economic exercise that India achieved. An insight into the factors that could have driven its economic growth would be largely helpful for public policy making in many developing countries across the world.

In this context, the present study aims at identifying the key determinants of economic growth in India over the period 1971–2016 by employing auto-regressive distributed lag (ARDL) bounds testing approach developed by Pesaran et al. (2001). This study distinguishes itself from the other studies, due to its focus on various sources of economic growth, including foreign, financial and real sources.

The rest of the article is structured as follows. Section II provides a brief history of Indian economic growth. The same section also includes a literary review on the factors driving India's economic growth. Section III describes the analytical framework of the present study followed by a discussion of issues related to data and estimation techniques in Section IV. Section V discusses the empirical results of the study. Lastly, Section VI summarizes the study and provides broad conclusions.

II. Indian Economic Growth: A Brief History

At the stroke of midnight, as India attained its independence from its colonial masters, it was one of the poorest economies in the world, faced with a challenge of feeding a huge population—largely illiterate and poor. On the other hand, the manufacturing sector was less productive and financial system was underdeveloped. It was a time when pessimists like Winston Churchill had foreseen that the country would disintegrate into chaos, given its complex and diverse economic, political and social structure. In the initial years of independence, India's policy favoured the planned system of development, influenced by socialist principles. Although this model worked in the initial years to take care of the needs of a poor state, it was not sufficient enough to attain higher growth rates. The principal hurdle was the large-scale regulation and control of the resources by the state. In fact, it led to inefficiencies in the system in general and the manufacturing sector in particular. The insulated trade and commerce and a tightly regulated financial system further compounded the problem. Thus, with a controlled trade, an over-regulated financial system and an underperforming manufacturing sector, India's first three decades after independence were marked by an abysmally low, average growth rate of 3.5% per annum. Although it touched 5.6% in the decade of 1980s, the political instability due to coalition politics and the underlying economic inefficiencies pushed the country towards the brink of failure by the end of the decade of 1980s.

The trigger came in the form of the Balance of Payments crisis of 1991. In fact, it was a culmination of India's policy failure over decades. This crisis eventually made India to embrace the economic reform path and there was no other option left out except to bring in structural reforms in the economy (Ahluwalia, 2002). These reform measures included trade and financial sector reforms. The financial system witnessed a large-scale overhaul, where the capital flows are allowed to flow freely, which in turn improved the efficiency of financial markets (Basu, 2009). The reform measures started yielding results. During the period between 2001 and 2008, India's economy grew at an average growth rate of 7.4% per annum. The average growth rate touched the peak of 8.4% during the period 2004 and 2008 and later declined to 7.4% during 2009–16. However, during 2014–18, India was the fastest-growing major economy in the world and at one stage in it even surpassed China in terms of growth rate. It is expected to surpass Japan by 2050, standing next only to the United States of America and China according to the *Lancet* journal.

This achievement was the logical corollary of the cautious and calibrated approach of successive governments and India's Central Bank that brought stability into the system, over a period of time through gradual, reformist approach, guided by carefully crafted and precisely executed policy decisions. On the front of deregulating financial markets, India was cautious. In fact, this approach helped a great deal in insulating the country's financial system from multiple financial crises, while the rest of the world suffered, and given the deeper integration of global financial markets (Tharoor, 2010). Indian financial system not only grew robust and became deeper but also played an important role in the country's economic growth. However, this economic growth was not even across sectors, in the post-reform India, due to the inherent, sector-specific challenges. While the service sector gave a major push to the country's growth prospects in the post-reform India, the agricultural sector's contribution to India's GDP kept falling. Even the agricultural incomes a percentage of GDP has fallen to 3.1% during 2012–18, from 4.4% during the period 2002–11. On the other hand, the value added in the agricultural sector as per cent of GDP 15.96% in 2019 from 27.33% in 1991. With nearly 42% of the country's total work force employed in the agriculture sector, its declining contribution to the GDP and the falling incomes raises alarm bells about its productivity and the need to address the structural and institutional challenges plaguing it. While it is pertinent to bring in reforms in the primary sector of the economy, it is equally important to revamp the Indian financial system that has been providing a scaffolding support to the country's growth story. Although it appears to be stable and strong, there is a need for another set of reforms, to make it robust enough to withstand the newer set of challenges that are in waiting, given the changing geo-political realities and their impact on global financial markets.

Literature Review on Factors Driving India's Economic Growth

Indian economy achieved a remarkable progress on its growth front and the pace and speed at which it occurred evinced interest in the determinants behind this achievement. As a result, a plenty of literature came into existence, and several dimensions and variables have been considered, in order to study the impact of various macroeconomic factors on India's economic growth. These studies can be classified broadly into two groups based on the factors considered and the analytical frameworks used. First group of studies (Dua & Rashid, 1998; Mallick & Behera, 2020; Sharma & Sahni, 2015) entirely focused on the relationship between individual factors and economic growth in a bivariate framework. These studies have largely sought to emphasize the role of individual factors such as inflation, human capital, foreign direct investment, trade openness and financial development in economic growth. The empirical results of these studies, ultimately, confirm that all such variables, both in the short run and long run, have a strong relationship with economic growth. In some cases, the evidence is, however, inconclusive regarding the direction of causality between the factors considered and economic growth. Dua and Rashid (1998), for instance, examined the role of foreign direct investment in India's economic growth during post-reform period using vector autoregressive model. The findings of the study show that FDI flows have significantly responded to economic activity in India during the study period. Similarly, Sharma and Sahn (2015) empirically tested the casual relationship between the investment on human capital and economic growth in India using annual data for the period 1991–92 to 2012–13. The evidence of the study confirms that a long-run relationship exists between human capital and economic growth in India.

On the other hand, there is another group of studies (Jain et al., 2015; Krishna, 2004; Sukla, 2020) which focused on the impact of group of macroeconomic variables on economic growth in a multivariate framework. Jain et al. (2015), for instance, investigate the impact of external factors on sector-wise GDP components in India for 2001–12 using multi-regression analysis. The results of this study indicate that foreign direct and institutional investment, imports have a significant impact on GDP components in India during the study period. More recently, Sukla (2020) investigated the impact of household consumption expenditure, interest rate, inflation and foreign investment flows on India's economic growth using annual data for 1978–2018. Findings from this study show that household consumption expenditure, interest rate, inflation and foreign investment flows have a significant impact on India's economic growth.

Despite a relatively vast body of literature on the determinants of the economic growth in the Indian context, there is still a lack of consensus on what drives India's economic growth. In this context, the present study is an attempt to provide a different perspective to this widely researched and debated issue. It considers the variables like per capita GDP growth rate, per capita capital, financial development, human capital development, inflation and trade openness and carries out the empirical exercise, to realize the objective of the study.

III. Analytical Framework

This study makes an attempt to the factors driving India's economic growth. Simple neoclassical production function is used here, where the level of output is considered to be a function of the stock of capital and the amount of labour put into use. Equation (1) depicts the simple production function of the neoclassical growth model:

$$Y = f\left(K, L\right) \tag{1}$$

In the above equation, Y is the aggregate real output produced, K is the capital stock and L is the amount of labour required to produce Y. Thus, it can be endorsed from the production function that, any increase in the variable, Y could come from one of the two sources. By multiplying both sides of Equation (1) with $1/L$, production function is derived, which shall be referred as per capita form.

$$\begin{aligned} \frac{Y}{L} &= f\left(\frac{K}{L}, \frac{L}{L}\right)(\text{or})\frac{Y}{L} = f\left(\frac{K}{L}, 1\right) \\ y &= f\left(k\right) \end{aligned} \tag{2}$$

In Equation (2), y denotes per capita output and k refers to per capita capital. The production function derived above implies that the stock of capital is one of the principal variables that affect the economic growth. Given its importance, the role of capital in fostering economic growth has been extensively reviewed, both theoretically and empirically, and documented in the literature. Early growth models (Domar, 1946; Harrod, 1939; Solow, 1956) have consigned a strategic role to capital accumulation in their respective growth models. Harrod–Domar growth model,[2] for example, argues that capital accumulation, that is, net investment would play a dual role in economic growth. While it generates income, on the one hand, it also increases the productivity of the economy on the other hand (Sato, 1964). The neoclassical growth theory also claims that the stock of capital plays a major role in economic growth along with labour supply and technological progress (Sato, 1964). Eventually, the significance given to the capital stock by these theories further gave way to a wide range of empirical studies like Mason (1988), Dollar (1991) and Mankiw et al. (1992). They attempted to investigate the association between investment and economic growth. However, the findings of the majority of these studies have been largely inconclusive.

In a similar fashion, human capital is another pertinent factor that has been studied extensively from the perspective of its contribution to economic growth. The endogenous growth models (Lucas, 1988; Romer, 1986) argued that human capital is one of the most important sources of economic growth. According to Romer (1986), human capital can be seen as 'knowledge' and 'ideas'. It has been further argued that it increases the total factor productivity and promotes economic growth, by enhancing the technology levels applied in the production process. Lucas (1988), in his endogenous growth model, also viewed human capital as one of the most significant factors of production. According to his growth model, human capital will not only be useful in producing final goods and services but also in accumulating more physical capital that will facilitate further production. Based on these theoretical beliefs, a large number of empirical studies (Barro, 1991; Hanushek & Wößmann, 2007; Pelinescu, 2015) have found evidence stating that human capital, measured in terms of healthy and educated population is a key determinant of economic growth.

Another important factor that many theoretical (Levine, 1997; Pagano, 1993; Schumpeter, 1911) and empirical scholars in the literature have highlighted as one of the major sources of economic growth is financial development. It is argued that financial development has a strong and independent role in increasing economic prosperity in the long run. For instance, Schumpeter (1911), argued that a well-functioning financial system in general and the active banking system, in particular, promotes economic growth. This could be achieved by the identification and funding of the entrepreneurs with fair chances of success and also by executing innovative production methods. The same is also evident from many empirical studies (Christopoulos & Tsionas, 2004; King & Levine, 1993; Lartey, 2010; Rajan & Zingales, 1998). Despite numerous studies, the role of financial development in economic growth has been the subject of longstanding debates among economists, as there are other empirical studies (Goldsmith, 1969; Liang & Teng, 2006; Shan et al., 2001) with divergent results.

In addition to this, other macroeconomic variables like trade liberalisation and inflation have also gained attention as possible determinants of economic growth. Inflation could possibly result in uncertainty over the profitability of investment projects in future. This may result in investors opting for traditional investment strategies, which eventually leads to low investment levels and lesser economic growth (Gokal & Hanif, 2004). Many empirical studies (Baltar, 2015; Fischer, 1993; Khan & Senhadji, 2000) have attempted to examine the impact of inflation on economic growth. Nevertheless, their results have been quite contradictory and divergent. Similarly, it has been observed that a country with more trade openness relatively outperforms its counterparts who are less open in trade (Edwards, 1998; Keho, 2017). The underlying reason for this claim is the potential of trade openness to enhance economic growth by exploiting the comparative advantage (Tyler, 1981). Another school of thought based on Prebisch–Singer Hypothesis (1949), argues that trade liberalisation in countries specialised in the production of low-quality products can negatively impact economic growth (Salvatore, 2010).

By considering the variables discussed above, the following production function is derived, in order to investigate the factors driving economic growth in India. It is represented in the following equation:

$$Y = f\left(PCK, FD, HCD, INFL, TOP\right) \tag{3}$$

where Y represents per capita GDP growth rate, PCK denotes per capita capital, FD refers to financial development, HCD is human capital development, $INFL$ represents inflation and TOP represents trade openness. Finally, the foregoing models are used for the empirical exercise of assessing the determinants of economic growth in India using the ARDL model.

IV. Data and Estimation Techniques

Annual data from the year, 1971–2016 has been used for the empirical analysis, covering a span of 46 years. Various sources have been used to collect data. Data for economic growth, inflation (*INFL*), per capita capital (*PCK*), trade openness (*TOP*) and financial development (*FD*) have been collected from the handbook of statistics on the Indian Economy published by the Reserve Bank of India (RBI). World Bank's report on world development indicators was used to collect the data on human capital development. While per capita GDP growth rate is used to measure the economic growth (*Y*), per capita capital is measured as the ratio of gross fixed capital formation to the size of the population. The present study also attempts to produce a broad-based index of financial development (FD). This index captures how well-developed, Indian financial markets and institutions have evolved in terms of their accessibility, depth and efficiency by taking the relevant financial proxies[3] into account. Over the past few years, numerous studies (Huang, 2011; Hye, 2011; Keho, 2010; Khan & Qayyum, 2006) have developed similar indices of financial development for different countries. However, in most of these studies, traditional measures were considered and efficiency parameters were completely ignored. Hye (2011), for instance, compiled a composite index of financial development for India by taking into account four indicators: liquidity liability ratio, commercial bank's private credit ratio, ratio of broad money (M3) to narrow money (M1) and the market capitalization ratio. No single efficiency parameter of financial development is used. The three variables (*Y, PCK* and *FD*) are transformed into logarithmic form as they are highly skewed variables in the given dataset. The variables are symbolised with the letter 'L' before each of them. While the school enrolment rate is used as a proxy for human capital development in India, the growth rate of a wholesale price index is taken as a measure of inflation. In addition to these, the trade openness is measured as the ratio of the total volume of foreign trade (export + imports) to GDP.

Econometric Specifications

Since variables used in the study are of a different order of integration [i.e. $I(0)$ and $I(1)$], the present study employed ARDL bounds testing. In comparison to other co-integration models (Engel & Granger, 1987; Johansen & Juselieus, 1990; Phillips & Hansen, 1990), ARDL bounds testing approach serves many other purposes. First, it can be applied even if the set of variables taken in the study are likely to be of a mix of $I(0)$ and $I(1)$ variables (Pesaran et al., 2001; Pesaran & Shin, 1999). Second, it performs much better and produces far better results than any other co-integration tests in small sample cases (Pesaran et al., 2001). Third, given the nature of inter-relationship between independent and dependent variables, the bounds testing approach to co-integration within the ARDL framework has been considered as a suitable method. This is due to the fact that it addresses the problem of residual serial correlation and also the potential endogeneity problem (Narayan & Smyth, 2006).

The presence of serial correlation, and drift and trends in the proposed series has been checked by using the Dickey–Fuller test (1979) and Phillips–Perron test (1988), respectively. Before estimating the ARDL model, the prerequisite exercise is to apply the bounds test to establish a long-term relationship between the variables included in the model (Pesaran et al., 2001). The error correction representation of the ARDL approach, based on Equation (3), is specified as

$$\Delta LY = \lambda_0 + \lambda_1 LY_{t-1} + \lambda_2 LPCK_{t-1} + \lambda_3 LFD_{t-1} + \lambda_4 HCD_{t-1} + \lambda_5 INFL_{t-1} + \lambda_6 TOP_{t-1} + \sum_{i=1}^{p}\beta_i \Delta LY_{t-i} + \sum_{j=0}^{q}\gamma_j \Delta LPCK_{t-j} + \sum_{k=0}^{r}\delta_k \Delta LFD_{t-k} + \sum_{l=0}^{s}\Omega_l \Delta HCD_{t-l} + \sum_{m=0}^{t}\theta_m \Delta INFL_{t-m} + \sum_{n=0}^{u}\sigma_n \Delta TOP_{t-n} + \varepsilon_t \tag{5}$$

The Wald test (F-statistics), derived from the above equation, forms pertinent part of the ARDL procedure. It helps to assess the existence of a long-run relationship among the model's variables. The Wald test (F-statistics) can be calculated by imposing linear restrictions on the estimated long-run coefficients of one-period lagged level of variables that have been taken into consideration. By testing the null hypotheses of no co-integration, against its alternative hypothesis of a co-integrating relationship, the existence of a long-run relationship among the variables can be established. The null hypotheses and alternative hypotheses for the Wald test are mentioned in the equation below:

H_0: $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = \lambda_5 = \lambda_6 = 0$ (no long-run relationship)
H_a: $\lambda_1 \neq \lambda_2 \neq \lambda_3 \neq \lambda_4 \neq \lambda_5 \neq \lambda_6 \neq 0$ (a long-run relationship exists)

The calculated F-test can be compared with the critical values given by Pesaran et al. (2001) and Narayan (2005) for the sake of testing hypothesis. According to these studies, the lower bound critical values assumed that the explanatory variables are integrated of order zero [$I(0)$]. On the other hand, the upper bound critical values assumed that the explanatory variables are integrated of order one [$I(1)$].

Hence, if the computed F-statistics is found to be smaller than the lower bound value, then the null hypothesis of no co-integration cannot be rejected. Converse to this, if the computed F-statistic is found to be greater than the upper bound value $I(1)$, then the null hypothesis of no co-integration can be rejected. However, if the computed F-statistic falls between the lower bound values and the upper bound values, then the results are inconclusive.

Once the co-integrating relationship is been detected successfully, the short-run and the long-run parameters of the co-integration equations can be estimated. The long-run co-integration relationship is estimated as mentioned below:

$$LY = \lambda_0 + \lambda_1 LY_{t-1} + \lambda_2 LPCK_{t-1} + \lambda_3 LFD_{t-1} + \lambda_4 HCD_{t-1} + \lambda_5 INFL_{t-1} + \lambda_6 TOP_{t-1} + v_t \quad (6)$$

Similarly, the short-run dynamic parameters may be obtained by estimating the VEC model as

$$\Delta LY = \sum_{i=j}^{p} \beta_i \Delta LY_{t-i} + \sum_{j=0}^{q} \gamma_j \Delta LPCK_{t-j} + \sum_{k=0}^{r} \delta_k \Delta LFD_{t-k} + \sum_{l=0}^{s} \Omega_l \Delta HCD_{t-l} + \sum_{m=0}^{t} \theta_m \Delta INFL_{t-m} + \sum_{n=0}^{u} \sigma_n \Delta TOP_{t-n} + \eta ECT_{t-1} + \varepsilon_t \quad (7)$$

where β, γ, δ, Ω, θ and σ are the short-run dynamic coefficients and η is the coefficient of the speed of adjustment which is expected to have a negative sign.

V. Results and Discussion

Given the tendency of the time-series data, to exhibit a non-stationary pattern, the Phillips–Perron test and Augmented Dickey–Fuller test have been used as a unit root test to check the stationarity properties of comprised variables, before testing their co-integration relationship. The results presented in Table 1

Table 1. Unit Root Test Results.

	ADF Test		PP Test		
Variables	Levels	First Diff.	Levels	First Diff.	Inference
LY	0.22	0.00*	0.26	0.00*	I(1)
LPCK	0.00*	–	0.00*	–	I(0)
LFD	0.19	0.00*	0.20	0.00*	I(1)
HCD	0.96	0.02**	0.92	0.01**	I(1)
INFL	0.00*	–	0.00*	–	I(0)
TOP	0.92	0.00*	0.89	0.00*	I(1)

Source: The author

Notes: (1) Based on P-value at the respective degrees of freedom, *, ** denote the level of significance at 1%, 5% and 10%, respectively.

show that variables including *INFL* and *LPCK* are found to be stationary at levels. On the other hand, all other remaining variables (*LFD, LY, TOP* and *HCD*) are found to be stationary at the first difference. Thus, a combination of $I(0)$ and $I(1)$ variables provided enough reason to use the ARDL bounds testing approach, in order to test the co-integration relationship between a set of selective macroeconomic variables and real per capita GDP growth in India.

In order to test the presence of a long-run relationship among the variables, the present study estimated the ordinary least squares for the first difference of both independent and the dependent variables. In the later stage, using the Wald test F-statistics, the study also examined the joint significance of parameters of the lagged level variables added to the regression. The computed F-statistic checks joint null hypothesis that the coefficients of the lagged level variables are zero (i.e. no long-run relationship exists between them). The empirical results of the bounds testing approach presented in Table 2 reveal that the computed F value is higher than the upper bound critical value of Pesaran et al. (2001) and Narayan (2004) (i.e. 6.41 > 5.64 or 4.43) at 1% level of significance, meaning that the null hypothesis of 'no co-integration' is rejected in favour of the alternative hypothesis of 'a co-integration relationship' at 1% level of significance, confirming the existence of a long-run association between dependent and a set of

Table 2. Results of Bounds Test (Unrestricted Intercept and no Trend).

Country	F-Statistics Value	Lag Length	Significance Level	Bound Critical Values by Narayan (2004)		Bound Critical Values by Pesaran (2001)	
				I(0)	I(1)	I(0)	I(1)
			1%	3.80	5.64	3.15	4.43
India	6.41*	1	5%	2.8	4.21	2.45	3.61
			10%	2.35	3.60	2.12	3.23

Source: The author

Notes: (1) Based on *P*-value at the respective degrees of freedom, denote the level of significance at 1%.

Table 3. Estimated Long-run Coefficients (Log of Real Per Capita GDP Growth Rate in period '*t*' $(ly)_t$ as Dependent Variable).

Regressor	Coefficient	Prob.
$(LPCK)_t$	0.59	0.02**
$(LFD)_t$	0.48	0.04**
$(HCD)_t$	0.02	0.21
$(INFL)_t$	−0.01	0.03**
$(TOP)_t$	−0.003	0.57
C_t	−0.23	0.36

Source: The author

Notes: (1) Based on *P* value at the respective degrees of freedom, ** denote the level of significance at 1%, 5% and 10%, respectively. (2) Null hypothesis: The series has a unit root.

independent variables. Subsequently, taking the real per capita growth rate as the dependent variable, the present study has also estimated the long-run coefficients.

The estimated results for the long-run coefficient presented in Table 3 indicate that *LFD, LPCK*, and *INFL* appear to have a significant (at 5% level of significance) impact on economic growth in the long run. However, the sign of the coefficient of *LFD* and *LPCK* appears to be positive, while it is negative for *INFL*. The significant positive impact of *LFD* and *LPCK* on economic growth implies that a 1% increase in the per capita capital (or) financial development leads to an increase in per capita GDP growth in India by 0.5% and 0.6%, respectively. These empirical results are in tandem with the studies of Schumpeter (1911), Pagano (1993) and Levine (1997). In the same way, the negative sign of inflation to economic growth indicates that, in the long run, an increase in the general price level will have a significant negative impact on India's real per capita GDP growth rate. On the other hand, the results suggest that the contribution of human capital and trade openness to India's economic growth is found to be negligible in the long run, as per the study period taken into consideration.

It can be inferred from the empirical results on error correction representation shown in Table 4 that real per capita GDP growth rate in India is positively influenced *HCD* and by *LPCK* at 5% and 1% level of significance, respectively, in the short run. Although *INFL* seems to have a significant impact (at 1%)

Table 4. Error Correction Representation.

Regressor	Coefficient	Prob.
ΔLPCK	1.11	0.00*
ΔLFD	0.01	0.68
ΔHCD	0.01	0.02**
ΔINFL	−0.002	0.00*
ΔTOP	−0.67	0.56
ECM_{t-1}	−0.21	0.00

Source: The author

Notes: (1) Based on *P* value at the respective degrees of freedom, *, ** denote the level of significance at 1%, 5% and 10%, respectively.

Table 5. Diagnostic Tests.

Diagnostic Tests	Test Statistics	*P* value	Test Results
Breusch–Godfrey's correlation LM test	χ^2 (1) = 0.93	0.33	No serial correlation (H_0: not rejected)
Breusch–Pagan–Godfrey's heteroscedasticity test	χ^2 (1) = 1.17	0.28	No heteroscedasticity
Jarque–Bera's normality test	χ^2 (2) = 0.62	0.62	Normal distribution
Ramsey's RESET test	χ^2 (1) = 2.92	0.10	Model is correctly specified
R^2 = 0.99, *R*-bar-square = 0.99, DW statistics: 1.72			

Source: The author

Notes: *, ** and *** denote the level of significance at 1%, 5% and 10%, respectively.

on real per capita GDP growth rate in India, like in the long run, its sign of a relationship with economic growth is found to be negative. The negative relationship between inflation and economic growth—both in the short and long run—may be due to the non-linear nature of the nexus between inflation and economic (Gillman & Kejak, 2000). On the other hand, in the short run, the contribution of trade openness and financial development towards economic growth has been found to be insignificant during the study period. Lastly, the short-run error correction term (ECM_{t-1}) is found to be negative at a 1% level of significance and also confirms the existence of a stable long-run association between regress and the regressors. Further, the negative sign of coefficient (−0.21) of the error correction term implies that deviation from the long-run real per capita GDP growth rate is corrected by 21% in the next period.

Thereafter, the present study conducted various diagnostic tests[4] to check the robustness of estimated empirical results obtained using the ARDL model. The results presented in Table 5 confirm that the model could not reject the null hypothesis (H_0) of no autocorrelation (χ^2 statistics = 0.93), no homoscedasticity (χ^2 statistics = 1.17) and the assumption of normality (χ^2 statistics = 0.62). Similarly, the empirical model is correctly specified as Ramsay's RESET test (F-statistics = 2.92) could not reject the null hypothesis of the linear specification model.

The stability of the estimated regression coefficients is evaluated using cumulative sum (CUSUM) and cumulative sum of squares (CUSUM SQ) tests. It is evident from Figures 1 and 2 that the plots of the CUSUM and CUSUM-Q are found to be within the boundaries with critical bounds at 5% level significance. This confirms the stability of the model's regression coefficients.

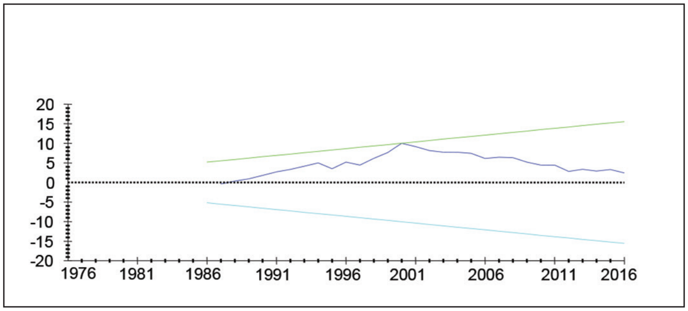

Figure 1. CUSUM Test.

Note: The straight lines represent critical bounds at 5% significance level.

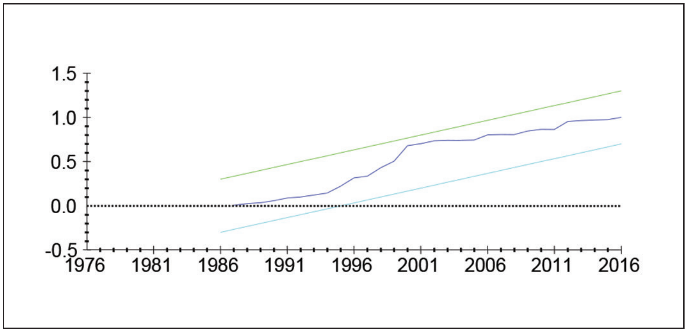

Figure 2. CUSUM-SQ Test.

Note: The straight lines represent critical bounds at 5% significance level.

VI. Concluding Remarks

India's rise from a state of abysmal growth rates to the status of being one of the fastest-growing economies of the world is not less than an economic miracle. Literature in the Indian context attributed this growth performance to a variety of factors. Still, as there is no consensus on the aspect of what drives India's economic growth, this study made an attempt to address this question, by taking into consideration, some real and financial variables. This combination makes it a different one from other studies. The present study uses the ARDL bounds testing model to empirically investigate the impact of selective macroeconomic variables on economic growth in India over the period 1971–2016. The long-run estimates of the ARDL model indicate that financial development and per capita capital have a significant positive impact on the country's economic growth. The results reiterated the fact that by channelising the savings towards investments that are productive, an active and a well-developed financial system accelerates economic growth.

As these findings hold important policy insights for India, it is pertinent to focus on the institutional mechanisms that further strengthen the Indian financial system which would increase its depth, scope and stability to foster economic growth. Similarly, the empirical results confirmed a negative impact of inflation on economic growth in India during the study period. It underscores the fact that an increase in the general price level reduces returns on labour efforts, resulting in people substituting from consumption to leisure (Greenwood & Huffman, 1987). Therefore, when the quantity of labour decreases in response to an increase in inflation, the return on capital also falls (Cooley & Hansen, 1989), resulting in a decline in the level of capital stock and output.

Another fallout of the higher inflation is the possibility that it could reduce the real wages of the labour. If not addressed, this could possibly become one of the factors that could contribute to the stagnation of real wages. With inflation remaining out of the bounds of the Reserve Bank of India for 10 months in 2022, it is pertinent to toughen the stance on controlling inflation in the first place, in tandem with the structural reforms that could boost productivity. Hence, the policymakers need to focus on strengthening anti-inflationary measures, through supply-side reforms, to avoid the negative long-run effects of inflation on economic growth. This study also suggests that in the short run, there is a positive association between economic growth and financial development, while the association of economic growth with trade openness has been negative. However, its relationship with both variables is found to be statistically insignificant. Contrary to the long run, the estimated results of error correction representation, for the short run, indicate that per capita capital along with human capital development has a significant positive impact on economic growth. This indicates the need for an emphasis on public and private investments in human capital, which further fosters economic growth.

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Notes

1. For instance, the economic growth in some developing economies in Asia has been relatively more rapid, than their Latin American counterparts during 1976–85. Per capita income during this period grew at an average

annual rate of 3.4% for 16 Asian economies. However, it declined at rates of 0.3% in 24 Latin American countries and 0.4% in 43 African countries (Dollar, 1992).

2. Harrod–Domar Model assumes a simple production function (i.e., $Y = \beta K$) with constant capital–output ratio (β). Therefore, any change in national output (ΔY) must be equal to β times ΔK, that is, $\Delta Y = \beta \Delta K$ representing capital as a major source of economic growth.
3. The proxies used in the construction of a summary measure of financial development are:

   $\text{Liquidity liability ratio} = \text{Broad money}\,(M3)/\text{Nominal GDP}$,

   $\text{Market capitalisation ratio} = \text{Value of shares listed at stock market/Nominal GDP}$,

   $\text{Return on assets} = \text{Net profit after tax/Assets}$,

   $\text{Turnover ratio} = \text{Value of equities trade/Value of shares listed at the stock market}$,

   $\text{Bank credit ratio} = \text{Total bank credit/Nominal GDP}$,
4. These diagnostic tests include Breusch–Godfrey's correlation LM test, Breusch–Pagan–Godfrey's heteroscedasticity test, Jarque–Bera's normality test and Ramsey's RESET test.

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